GMM and BLP Estimation

HKUST(GZ) Structural Estimation Reading Group

Oct 12 2026

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Introduction

The slides materials are from

- ResEcon 703 course at the University of Massachusetts Amherst, by Matt Woerman
- Berry, Levinsohn, and Pakes (1995)

Agenda

GMM

- Method of moments overview
- Method of moments estimator
- Method of moments examples
- GMM estimator
- GMM example
- Properties of the GMM estimator
- Optimal GMM estimator
- Specification and hypothesis tests
- Logit model with GMM

BLP Estimation

- Endogeneity in Structural Models
- BLP Estimation
- BLP 1995 Paper Reading

Method of Moments Overview

(Generalized) Method of Moments

Method of moments (MM) estimation is a special case of generalized method of moments (GMM) estimation

- Common in modern empirical economics
  - Most common in industrial organization, macroeconomics, and finance
- Less parametric than maximum likelihood
  - Requires assumptions about moment conditions, not assumptions about the full distribution of data
- Most of the estimators you have learned so far are (G)MM estimators
  - OLS, 2SLS, GLS, MLE, etc.

Overview of (generalized) method of moments estimation

- MM and GMM use population moment conditions to estimate parameters
  - Moments can come from economic models, statistical assumptions, etc.
- MM and GMM estimators "solve" the sample moment conditions that are analogous to these population moment conditions

Moment Conditions

Moment conditions are functions of parameters and data that equal zero in expectation when evaluated at the true parameter values

$$\begin{aligned}
\text{Mean:} \quad & \mu_y = E[y] && \Rightarrow E[y - \mu_y] = 0 \\
\text{Var:} \quad & \sigma_y^2 = E[(y - \mu_y)^2] && \Rightarrow E[(y - \mu_y)^2 - \sigma_y^2] = 0 \\
\text{Cov:} \quad & \sigma_{xy} = E[(y - \mu_y)(x - \mu_x)] && \Rightarrow E[(y - \mu_y)(x - \mu_x) - \sigma_{xy}] = 0
\end{aligned}$$

How do we normally generate moment conditions?

- Economic models
  - Example: first-order conditions
- Econometric or statistical model
  - Example: instruments must be uncorrelated with errors
- Model fit
  - Example: predicted market shares must equal realized market shares

Method of Moments Intuition

Suppose we have five random draws, but we do not know the distribution

$$\boldsymbol{y} = \{5, 10, 9, 14, 7\}$$

Suppose the mean of the unknown distribution is given by μ, which gives us a population moment condition

$$E[y_i] = \mu \quad \Rightarrow \quad E[y_i - \mu] = 0$$

If this moment conditions holds in the population, then we should expect an analogous condition to hold in our sample

$$\frac{1}{n} \sum_{i=1}^{n} (y_i - \mu) = 0$$

We can estimate the population parameter, $\widehat{\mu}$, by finding the value that makes this empirical expression true in our sample

Method of Moments Estimator

Method of Moments Assumptions

Data, $\{y, \boldsymbol{x}, \boldsymbol{z}\}$, are drawn from a population with K moment conditions that are functions of K parameters, $\boldsymbol{\theta}$

$$E[\boldsymbol{m}(y, \boldsymbol{x}, \boldsymbol{z}, \boldsymbol{\theta})] = \mathbf{0}$$

- $\boldsymbol{m}(\cdot)$: vector of K functions
- $\boldsymbol{\theta}$: vector of K parameters
- y: dependent variable
- $\boldsymbol{x}$: vector of independent variables
- $\boldsymbol{z}$: vector of exogenous instruments

We make assumptions about the moment conditions of the population from which the data are drawn, but we do not have to make an assumption about the full underlying distribution

Sample Moments

These moment conditions hold for the population from which the data are drawn

$$E[\boldsymbol{m}(y, \boldsymbol{x}, \boldsymbol{z}, \theta)] = \mathbf{0}$$

But we do not observe the full population, only a sample

We replace the population expectation with its empirical analog—the sample mean—to generate the corresponding K sample moments as a function of the K parameters

$$\frac{1}{n} \sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \theta) = \mathbf{0}$$

Method of Moments Estimator

The method of moments estimators is the set of parameters, $\widehat{\boldsymbol{\theta}}$, that solves the system of equation given by the sample moments

$$\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widehat{\boldsymbol{\theta}}) = \mathbf{0}$$

In some case, $\widehat{\boldsymbol{\theta}}$ will have a closed-form expression, but in other cases we can reformulate $\widehat{\boldsymbol{\theta}}$ as the solution to a minimization problem and use numerical optimization

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, Q(\boldsymbol{\theta})$$

where $Q(\boldsymbol{\theta})$ is given by

$$Q(\boldsymbol{\theta}) = \left[\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]' \left[\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]$$

Method of Moments Examples

Method of Moments Population Mean Example

We have five data points from a population, but what is the mean, μ, of this population?

$$\boldsymbol{y} = \{5, 10, 9, 14, 7\}$$

If y is i.i.d. with mean μ, then a population moment condition is

$$E[y - \mu] = 0$$

Replacing the expectation with the sample analog gives

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{\mu}) = 0$$

Solving this one equation for the one parameter yields

$$\widehat{\mu} = \frac{1}{n}\sum_{i=1}^{n} y_i = 9$$

Method of Moments OLS Regression Example

The OLS regression estimator is a special case of a MM estimator

Consider the general OLS regression model

$$y_i = \beta' \boldsymbol{x}_i + \varepsilon_i$$

- β: vector of K parameters
- $\boldsymbol{x}_i$: vector of K variables

We assume the error term, ε_i, is orthogonal to the data, $\boldsymbol{x}_i$

$$E[\boldsymbol{x}_i \varepsilon_i] = 0$$

Replacing ε_i with $y_i - \beta' \boldsymbol{x}_i$ gives K population moment conditions

$$E[\boldsymbol{x}_i (y_i - \beta' \boldsymbol{x}_i)] = \mathbf{0}$$

Method of Moments OLS Regression Example

We have K moments that are functions of the K parameters, β

$$E[\boldsymbol{x}_i(y_i - \beta'\boldsymbol{x}_i)] = \mathbf{0}$$

We replace these K expectations with the K sample analogs

$$\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i(y_i - \widehat{\beta}'\boldsymbol{x}_i) = \mathbf{0}$$

Solving these K equations for the K parameters yields the method of moments estimator

$$\widehat{\beta} = \left(\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{x}_i'\right)^{-1}\left(\sum_{i=1}^{n}\boldsymbol{x}_i y_i\right)$$

This method of moments estimator is equivalent to the traditional OLS estimator

Method of Moments Maximum Likelihood Example

The MLE can also be motivated as a special case of a MM estimator

Consider a random variable y with conditional density

$$f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})$$

Then the log-likelihood function of $\boldsymbol{\theta}$ conditional on $\boldsymbol{y}$ and $\boldsymbol{X}$ is

$$\ln L(\boldsymbol{\theta} \mid \boldsymbol{y}, \boldsymbol{X}) = \sum_{i=1}^{n} \ln f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})$$

Maximizing this log-likelihood function yields K first-order conditions, one for each of the K parameters

$$\mathbf{0} = \frac{\partial \ln L(\boldsymbol{\theta} \mid \boldsymbol{y}, \boldsymbol{X})}{\partial \boldsymbol{\theta}} = \sum_{i=1}^{n} \frac{\partial \ln f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$$

Method of Moments Maximum Likelihood Example

From these K first-order conditions, we have that

$$\sum_{i=1}^{n} \frac{\partial \ln f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}$$

Dividing by n yields an expression that looks like sample moments

$$\frac{1}{n} \sum_{i=1}^{n} \frac{\partial \ln f(y_i \mid \boldsymbol{x}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{0}$$

These K sample moments can be viewed as the empirical analogs of K population moment conditions

$$E\left[\frac{\partial \ln f(y \mid \boldsymbol{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0}$$

MLE can be motivated as a MM estimator with these population moments

GMM Estimator

Generalized Method of Moments Assumptions

Data, $\{y, \boldsymbol{x}, \boldsymbol{z}\}$, are drawn from a population with L moment conditions that are functions of K parameters, $\boldsymbol{\theta}$, with $L \geq K$

$$E[\boldsymbol{m}(y, \boldsymbol{x}, \boldsymbol{z}, \boldsymbol{\theta})] = \mathbf{0}$$

- $\boldsymbol{m}(\cdot)$: vector of L functions
- $\boldsymbol{\theta}$: vector of K parameters
- y: dependent variable
- $\boldsymbol{x}$: vector of independent variables
- $\boldsymbol{z}$: vector of exogenous instruments

We make assumptions about the moment conditions of the population from which the data are drawn, but we do not have to make an assumption about the full underlying distribution

Source of Moment Conditions

Generalized method of moments allows for more moment conditions than the number of parameters

- Method of moments requires the number of moment conditions to equal the number of parameters

Additional moment conditions can come from many sources

- Economic model: Demand parameter appears in demand and supply first-order conditions
- Instruments: Multiple instruments for one endogenous variable
- Model fit: Predicted market shares equal realized market shares
- Statistical assumptions: Errors are orthogonal to data

Sample Moments

These L moment conditions hold for the population from which the data are drawn

$$E[\boldsymbol{m}(y, \boldsymbol{x}, \boldsymbol{z}, \theta)] = \mathbf{0}$$

But we do not observe the full population, only a sample

We replace the population expectation with is empirical analog—the sample mean—to generate the corresponding L sample moments as a function of the K parameters

$$\frac{1}{n} \sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \theta) = \mathbf{0}$$

Generalized Method of Moments Estimator

With only K parameters, we cannot solve for L unique sample moments when $L > K$

- Instead, we will seek to simultaneously get as close as possible to solving all L sample moments

The generalized method of moments estimators is the set of parameters, $\widehat{\boldsymbol{\theta}}$, that minimizes the weighted sum of squared sample moments

$$\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\, Q(\boldsymbol{\theta})$$

where $Q(\boldsymbol{\theta})$ is given by

$$Q(\boldsymbol{\theta}) = \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]' \boldsymbol{W} \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]$$

and $\boldsymbol{W}$ is a $L \times L$ positive definite weighting matrix

GMM Example

Generalized Method of Moments 2SLS Regression Example

The 2SLS regression estimator is a special case of a GMM estimator

Consider the linear regression model

$$y_i = \beta' \boldsymbol{x}_i + \varepsilon_i$$

- β: vector of K parameters
- $\boldsymbol{x}_i$: vector of K variables, but some are endogenous

We have a vector of L instruments, $\boldsymbol{z}_i$, that are correlated with $\boldsymbol{x}_i$ but are orthogonal to the error term, ε_i

$$E[\boldsymbol{z}_i \varepsilon_i] = 0$$

Replacing ε_i with $y_i - \beta' \boldsymbol{x}_i$ gives L population moment conditions

$$E[\boldsymbol{z}_i (y_i - \beta' \boldsymbol{x}_i)] = \mathbf{0}$$

Generalized Method of Moments 2SLS Regression Example

We have L moments that are functions of the K parameters, β

$$E[\boldsymbol{z}_i(y_i - \beta'\boldsymbol{x}_i)] = \mathbf{0}$$

We replace these L expectations with the L sample analogs

$$\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i(y_i - \beta'\boldsymbol{x}_i) = \mathbf{0}$$

We cannot solve all L sample moment conditions with only K parameters (if $L > K$), so instead we find the set of parameters, $\widehat{\beta}$, that minimizes the weighted sum of squared sample moments

$$Q(\beta) = \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i(y_i - \beta'\boldsymbol{x}_i)\right]' \boldsymbol{W} \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i(y_i - \beta'\boldsymbol{x}_i)\right]$$

Generalized Method of Moments 2SLS Regression Example

The GMM estimator for this instrumental variables regression, $\widehat{\beta}$, is

$$\widehat{\beta} = \underset{\beta}{\operatorname{argmin}}\, Q(\beta)$$

where $Q(\beta)$ is given by

$$Q(\beta) = \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i(y_i - \widehat{\beta}'\boldsymbol{x}_i)\right]' \boldsymbol{W} \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{z}_i(y_i - \widehat{\beta}'\boldsymbol{x}_i)\right]$$

The GMM estimator will be equivalent to the 2SLS estimator if we use a weighting matrix, $\boldsymbol{W}$, of

$$\boldsymbol{W} = \sum_{i=1}^{n} \boldsymbol{z}_i \boldsymbol{z}_i'$$

But another weighting matrix might yield a "better" estimator

Properties of the GMM Estimator

Asymptotic Properties of GMM Estimator

If some assumptions about the empirical moments and the weighting matrix hold, the generalized method of moments estimator has these properties

1. Consistency
2. Asymptotic normality

Note that the GMM estimator is consistent and asymptotically normal, but it is not efficient, in contrast to the MLE

- The distributional assumption of maximum likelihood provides additional information that makes the MLE efficient

GMM Empirical Moment Assumptions

In these assumptions and properties, the 0 subscript denotes an object evaluated at the true parameter values, $\boldsymbol{\theta}_0$

1. The empirical moments obey the law of large numbers

$$\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta}_0) \xrightarrow{p} \mathbf{0}$$

2. The derivatives of the empirical moments converge

$$\frac{1}{n}\sum_{i=1}^{n} \frac{\partial \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\bigg|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \xrightarrow{p} \boldsymbol{G}_0$$

GMM Empirical Moment Assumptions

3. The empirical moments obey the central limit theorem

$$\frac{\sqrt{n}}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i,\boldsymbol{x}_i,\boldsymbol{z}_i,\boldsymbol{\theta}_0)\xrightarrow{d}\mathcal{N}(0,\boldsymbol{S}_0)$$

where

$$\boldsymbol{S}_0=\text{plim}\,\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n}\boldsymbol{m}(y_i,\boldsymbol{x}_i,\boldsymbol{z}_i,\boldsymbol{\theta}_0)\boldsymbol{m}(y_j,\boldsymbol{x}_j,\boldsymbol{z}_j,\boldsymbol{\theta}_0)'$$

4. The parameters are identified

$$\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i,\boldsymbol{x}_i,\boldsymbol{z}_i,\boldsymbol{\theta}_1)=\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{m}(y_i,\boldsymbol{x}_i,\boldsymbol{z}_i,\boldsymbol{\theta}_2)\quad\Leftrightarrow\quad\boldsymbol{\theta}_1=\boldsymbol{\theta}_2$$

GMM Weighting Matrix Assumption

5. The weighting matrix converges to a finite symmetric positive definite matrix

$$\boldsymbol{W} \xrightarrow{p} \boldsymbol{W}_0$$

Consistency of the GMM Estimator

The GMM estimator, $\widehat{\boldsymbol{\theta}}$, is a consistent estimator of the true parameter values, $\boldsymbol{\theta}_0$

$$\widehat{\boldsymbol{\theta}} \xrightarrow{p} \boldsymbol{\theta}_0$$

- As our sample size grows (to infinity), the GMM estimator becomes vanishingly close to the true parameter value(s)

Asymptotic Normality of the GMM Estimator

The GMM estimator, $\widehat{\theta}$, is asymptotically normal with mean θ_0 and known variance

$$\widehat{\theta} \overset{a}{\sim} \mathcal{N}\left(\theta_0, \frac{1}{n}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{S}_0\boldsymbol{W}_0\boldsymbol{G}_0)(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}\right)$$

where

$$\begin{aligned}
\boldsymbol{G}_0 &= \text{plim}\, \frac{1}{n}\sum_{i=1}^{n} \left.\frac{\partial \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} \\
\boldsymbol{W}_0 &= \text{plim}\, \boldsymbol{W} \\
\boldsymbol{S}_0 &= \text{plim}\, \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta}_0)\boldsymbol{m}(y_j, \boldsymbol{x}_j, \boldsymbol{z}_j, \boldsymbol{\theta}_0)'
\end{aligned}$$

Optimal GMM Estimator

GMM Variance-Covariance Matrix

Every weighting matrix yields a GMM estimator that is consistent

- But the choice of weighting matrix, $\boldsymbol{W}$, will affect the variance of the GMM estimator, $\widehat{\boldsymbol{\theta}}$, because its probability limit, $\boldsymbol{W}_0$, enters the calculation of the variance-covariance matrix

$$Var(\widehat{\boldsymbol{\theta}}) = \frac{1}{n}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{S}_0\boldsymbol{W}_0\boldsymbol{G}_0)(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}$$

If every weighting matrix yields a GMM estimator that is consistent, then we would ideally like to use the weighting matrix that minimizes the variance of the estimator

Optimal Weighting Matrix

$$Var(\widehat{\theta}) = \frac{1}{n}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{S}_0\boldsymbol{W}_0\boldsymbol{G}_0)(\boldsymbol{G}_0'\boldsymbol{W}_0\boldsymbol{G}_0)^{-1}$$

The variance of the GMM estimator is minimized when

$$\boldsymbol{W}_0 = \boldsymbol{S}_0^{-1} = \left[\text{plim}\,\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n}\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \theta_0)\boldsymbol{m}(y_j, \boldsymbol{x}_j, \boldsymbol{z}_j, \theta_0)'\right]^{-1}$$

which gives a variance-covariance matrix of

$$Var(\widehat{\theta}) = \frac{1}{n}(\boldsymbol{G}_0'\boldsymbol{S}_0^{-1}\boldsymbol{G}_0)^{-1}$$

$\boldsymbol{S}_0$ is the variance-covariance matrix of the empirical moments

- We minimize the variance of the GMM estimator by weighting each moment inversely to its variance

Optimal (or Two-Step) GMM Estimator

We want to use a weighting matrix, $\boldsymbol{W}$, that converges in probability to the variance of the empirical moments, $\boldsymbol{S}_0^{-1}$

- But we do not know $\boldsymbol{S}_0^{-1}$ because we do not know the true parameter values, $\boldsymbol{\theta}_0$

We can use a two-step procedure, performing GMM estimation twice, to obtain this optimal GMM estimator

1. Step 1: Perform GMM with any arbitrary weighting matrix and estimate the variance of the moments, $\widetilde{\boldsymbol{S}}$, using this GMM estimator
2. Step 2: Perform GMM again with $\boldsymbol{W} = \widetilde{\boldsymbol{S}}^{-1}$

This procedure is a generalization of the FGLS regression

We will use the `gmm()` function from the `gmm` package in R to find this optimal GMM estimator

Optimal GMM Estimator: Step 1

Step 1: Perform GMM with any arbitrary weighting matrix to obtain the first-step GMM estimator, $\widetilde{\boldsymbol{\theta}}$

- The GMM estimator is consistent for any weighting matrix, so we can use any arbitrary weighting matrix to obtain a consistent estimate of the true parameter values, $\boldsymbol{\theta}_0$
- The identity matrix—equally weighting each moment—is a simple one to use

Use this first-step GMM estimator, $\widetilde{\boldsymbol{\theta}}$, to construct an estimator of $\boldsymbol{S}_0$

- Also assume observations are independent to simplify this estimator

$$\widetilde{\boldsymbol{S}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widetilde{\boldsymbol{\theta}})\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widetilde{\boldsymbol{\theta}})'$$

Then use the inverse of this estimate, $\widetilde{\boldsymbol{S}}$, as the weighting matrix in the second step

Optimal GMM Estimator: Step 2

Step 2: Perform GMM with weighting matrix $\boldsymbol{W} = \widetilde{\boldsymbol{S}}^{-1}$

Find the GMM estimator, $\widehat{\theta}$, that minimizes the objective function

$$Q(\boldsymbol{\theta}) = \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]' \widetilde{\boldsymbol{S}}^{-1} \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})\right]$$

This second-step GMM estimator, $\widehat{\theta}$, also gives a consistent estimate of the true parameter values, $\boldsymbol{\theta}_0$, and it has the minimum variance among all GMM estimators

- It is not efficient (when compared to MLE), but it is efficient among GMM estimators

This optimal GMM estimator has the variance-covariance matrix

$$Var(\widehat{\boldsymbol{\theta}}) = \frac{1}{n}(\boldsymbol{G}_0' \boldsymbol{S}_0^{-1} \boldsymbol{G}_0)^{-1}$$

Optimal GMM Variance Estimator

The variance-covariance matrix of the GMM estimator is evaluated at the true parameter values, $\boldsymbol{\theta}_0$, which we do not know

- We construct an estimator of this variance-covariance matrix by evaluating at the optimal GMM estimator, $\widehat{\boldsymbol{\theta}}$

$$\widehat{Var}(\widehat{\boldsymbol{\theta}}) = \frac{1}{n}\left(\widehat{\boldsymbol{G}}'\widehat{\boldsymbol{S}}^{-1}\widehat{\boldsymbol{G}}\right)^{-1}$$

where the components of this matrix are estimated by

$$\widehat{\boldsymbol{G}} = \frac{1}{n}\sum_{i=1}^{n} \left.\frac{\partial \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\right|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}}$$

$$\widehat{\boldsymbol{S}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widehat{\boldsymbol{\theta}})\boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widehat{\boldsymbol{\theta}})'$$

Specification and Hypothesis Tests

Overidentifying Restrictions Test

When we have more moment conditions than parameters, the model is "overidentified" and we cannot ensure all moments equal zero simultaneously, but we can test if all moment conditions are sufficiently close to zero

$$H_0 : E[\boldsymbol{m}(y, \boldsymbol{x}, \boldsymbol{z}, \boldsymbol{\theta}_0)] = \mathbf{0}$$

When $\widehat{\boldsymbol{\theta}}$ is estimated by optimal GMM, the overidentifying restrictions test statistic is equal to the objective function evaluated at the optimal GMM estimator, $Q(\widehat{\boldsymbol{\theta}})$

$$OIR = \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widehat{\boldsymbol{\theta}})\right]' \widetilde{\boldsymbol{S}}^{-1} \left[\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{m}(y_i, \boldsymbol{x}_i, \boldsymbol{z}_i, \widehat{\boldsymbol{\theta}})\right]$$

which is asymptotically distributed χ^2 with $L - K$ degrees of freedom

$$OIR \overset{a}{\sim} \chi^2(L - K)$$

Hypothesis Tests

We discussed three hypothesis testing procedures for use with maximum likelihood estimation

- Likelihood ratio test
- Wald test
- Lagrange multiplier test

Each of these test procedures has an analogous test using the results of GMM estimation

- The intuition of each test is the same as its MLE counterpart, but using the GMM objective function is place of the likelihood or log-likelihood function

Refer to an econometrics textbook for the details of these hypothesis tests

Logit Model with GMM

Logit Model First-Order Conditions

When estimating the parameters of the logit model by maximum likelihood, the first-order conditions for the MLE are

$$\frac{\partial \ln L(\boldsymbol{\theta} \mid \boldsymbol{y}, \boldsymbol{X})}{\partial \boldsymbol{\theta}} = \mathbf{0}$$

where the log-likelihood function is given by

$$\ln L(\boldsymbol{\theta} \mid \boldsymbol{y}, \boldsymbol{X}) = \sum_{n=1}^{N} \sum_{i=1}^{J} y_{ni} \ln P_{ni}(\boldsymbol{x}_n, \boldsymbol{\theta})$$

We can alternatively express these first-order conditions as sample moments conditions

- See the Train textbook for the proof of this equality

Logit Model Moment Conditions

If we assume representative utility is linear, $V_{ni} = \beta' \boldsymbol{x}_{ni}$, then the logit MLE first-order conditions are equivalent to

$$\frac{1}{NJ} \sum_{n=1}^{N} \sum_{i=1}^{J} \left(y_{ni} - P_{ni}(\boldsymbol{x}_n, \beta) \right) \boldsymbol{x}_{ni} = \mathbf{0}$$

These sample moments can be viewed as the empirical analogs of the population moment conditions

$$E\left[\left(y_{ni} - P_{ni}(\boldsymbol{x}_n, \beta) \right) \boldsymbol{x}_{ni} \right] = \mathbf{0}$$

These population moment conditions state that the data, $\boldsymbol{x}_n$, are orthogonal to the model residuals, $y_{ni} - P_{ni}(\boldsymbol{x}_{ni}, \beta)$

- Analogous to the data and errors being orthogonal in OLS regressions

Logit Model with Method of Moments

If this assumption of exogeneity holds, then we can also estimate the parameters of the logit model by MM using the sample moments

$$\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\widehat{\beta})\right)\boldsymbol{x}_{ni}=\mathbf{0}$$

- MM because we have the same number of moments and parameters

Because logit choice probabilities are nonlinear, this MM estimator does not have a closed-form solution

- Instead, we will use numerical optimization to find the set of parameters, $\widehat{\beta}$, that minimizes the MM objective function

$$Q(\beta)=\left[\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\beta)\right)\boldsymbol{x}_{ni}\right]'\left[\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\beta)\right)\boldsymbol{x}_{ni}\right]$$

Logit Model Moment Conditions with Instruments

If the assumption of exogeneity does not hold, but we instead have some exogenous instruments, $\boldsymbol{Z}$, that are correlated with our data, $\boldsymbol{X}$, then we can create a different set of sample moment conditions that must hold

Our exogenous instruments, $\boldsymbol{Z}$, are orthogonal to the model residuals

$$E\left[\left(y_{ni} - P_{ni}(\boldsymbol{x}_n, \beta)\right) \boldsymbol{z}_{ni}\right] = \mathbf{0}$$

Replacing the expectation with the sample analog gives

$$\frac{1}{NJ} \sum_{n=1}^{N} \sum_{i=1}^{J} \left(y_{ni} - P_{ni}(\boldsymbol{x}_n, \widehat{\beta})\right) \boldsymbol{z}_{ni} = \mathbf{0}$$

Our instruments, $\boldsymbol{Z}$, can be any exogenous variables that are correlated with our data, $\boldsymbol{X}$, including the exogenous elements of $\boldsymbol{X}$

- We must have at least as many instruments (and moments) as parameters in order to identify all parameters

Logit Model with GMM

We can estimate the parameters of the logit model with instruments by GMM using the instrument sample moments

$$\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\widehat{\beta})\right)\boldsymbol{z}_{ni}=\mathbf{0}$$

We will use the two-step procedure described earlier—based on this objective function—to find the optimal GMM estimator

$$Q(\beta)=\left[\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\beta)\right)\boldsymbol{z}_{ni}\right]'\boldsymbol{W}\left[\frac{1}{NJ}\sum_{n=1}^{N}\sum_{i=1}^{J}\left(y_{ni}-P_{ni}(\boldsymbol{x}_n,\beta)\right)\boldsymbol{z}_{ni}\right]$$

1. Estimate $\widetilde{\beta}$ using any arbitrary weighting matrix in order to estimate $\widetilde{\boldsymbol{S}}$ and find the optimal weighting matrix
2. Use that weighting matrix to find the optimal GMM estimator, $\widehat{\beta}$

Endogeneity in Structural Models

Endogeneity in Structural Models

So far, we have (mostly) assumed that all of our explanatory variables are exogenous

- When we talked about GMM estimation, we talked how we can use it to incorporate instruments, but I did not say much about why we would want to do so

Why is exogeneity/endogeneity so important?

- We need exogenous variation in our explanatory variables in order to give our parameter estimates a "causal" interpretation
  - If the data are endogenous, our parameters can be interpreted as a kind of correlation between the data and choices, but they will not be the true structural parameters we intend to estimate
- But in most cases, exogenous variation in the explanatory variables is difficult to come by

Examples of Endogeneity

Housing choice and commute choice are correlated

- Example: people who like public transit tend to live closer to transit stations, making their transit travel time lower
- The coefficient on transit travel time will be biased upward

Price and unobserved quality are correlated

- Example: products with higher unobserved (to the econometrician) quality cost more and are preferred by consumers
- The coefficient on price will be biased downward and may even have the wrong sign

Price and unobserved marketing are correlated

- Example: large marketing campaigns may be accompanied by sales or increased prices
- The coefficient on price will be biased, but the direction is uncertain

Exogenous Variation and Causal Parameters

How do we estimate parameters with a "causal" interpretation?

- BLP estimation: use instruments to isolate exogenous variation in explanatory variables
- Control function estimation: use instruments to control for endogeneity in explanatory variables

What makes a good instrument?

- Correlated with explanatory variables
- Exogenous, or uncorrelated with random utility

Where do we get good instruments?

- Same concept as good research design for reduced-form analysis
  - Institutional knowledge, natural experiments, etc.

BLP Estimation

Demand Estimation for Differentiated Products

BLP (1995) develops a random coefficient demand model for differentiated products.
The key challenge is that products have unobserved characteristics:

$$\xi_j$$

which are observed by consumers and firms but not by researchers.

A demand equation must address:

$$Cov(p_j, \xi_j) \neq 0$$

because prices are endogenous.

Baseline Utility Specification

A special case of the BLP model:

$$U_{ij} = x_j\beta - \alpha p_j + \xi_j + \epsilon_{ij}$$

where:

- x_j: observed product characteristics
- p_j: price
- ξ_j: unobserved product characteristics
- ϵ_{ij}: consumer-product idiosyncratic shock

Define mean utility:

$$\boxed{\delta_j = x_j\beta - \alpha p_j + \xi_j}$$

Therefore:

$$U_{ij} = \delta_j + \epsilon_{ij}$$

Market Share in the Logit Model

Consumers choose product j if:

$$U_{ij} > U_{iq}, \quad \forall q \neq j$$

The market share is:

$$s_j = \int P_j(\delta_j - \delta_q + \epsilon) dP(\epsilon)$$

Under the i.i.d. extreme value assumption:

$$s_j = \frac{\exp(\delta_j)}{1 + \sum_q \exp(\delta_q)}$$

which has a closed-form solution.

Problem with Simple Logit

The simple logit model assumes:

$$U_{ij} = \delta_j + \epsilon_{ij}$$

where utility is additively separable.

This implies:

- substitution patterns depend only on market shares
- similar products are not necessarily closer substitutes
- cross-price elasticities are restricted

BLP introduces random coefficients to allow richer substitution patterns.

Random Coefficient Utility Model

BLP extends the model by allowing consumer heterogeneity:

$$U_{ij} = x_j\bar{\beta} - \alpha p_j + \xi_j + \sum_k \sigma_k x_{jk} v_{ik} + \epsilon_{ij}$$

where:

- $\bar{\beta}$: mean taste parameters
- σ_k: standard deviations of random coefficients
- v_{ik}: consumer-specific taste shocks

Consumers differ in their valuation of product characteristics.

Mean Utility and Consumer Heterogeneity

The random coefficient model can be written as:

$$U_{ij} = \delta_j + \mu_{ij}$$

where:

$$\boxed{\delta_j = x_j \bar{\beta} - \alpha p_j + \xi_j}$$

is the mean utility of product j.

The deviation from mean utility:

$$\boxed{\mu_{ij} = \sum_k \sigma_k x_{jk} v_{ik} + \epsilon_{ij}}$$

Random Coefficients and Market Shares

With consumer–product interactions, BLP computes market shares in two stages.

First, conditional on consumer characteristics v_i, integrate out the i.i.d. extreme-value shock ϵ_{ij}:

$$\boxed{f_j(v_i, \boldsymbol{\delta}, \boldsymbol{p}, \boldsymbol{x}, \boldsymbol{\theta}) = \frac{\exp[\delta_j + \mu_j(x_j, p_j, v_i; \theta_2)]}{1 + \sum_{r=1}^{J} \exp[\delta_r + \mu_r(x_r, p_r, v_i; \theta_2)]}} \tag{6.6}$$

Second, integrate over the distribution of consumer characteristics:

$$\boxed{s_j(\boldsymbol{p}, \boldsymbol{x}, \boldsymbol{\xi}, \boldsymbol{\theta}, P_0) = \int f_j(v_i, \boldsymbol{\delta}(\boldsymbol{x}, \boldsymbol{p}, \boldsymbol{\xi}), \boldsymbol{p}, \boldsymbol{x}, \boldsymbol{\theta})\, P_0(dv_i)} \tag{6.7}$$

- Equation (6.6) has a closed form conditional on v_i.
- Equation (6.7) generally has no closed form and is evaluated by simulation.

Contraction Mapping

This estimation framework is theoretically feasible

- But we have to estimate δ_{jm} for each product and market, which can easily be 100s or 1000s (or more!) of terms to estimate

BLP developed an alternate approach that does not require estimating these δ_{jm} terms in the standard way

- Their insight is that these δ_{jm} terms determine predicted market shares, so we want to find the set of constant terms that equates predicted market shares with observed market shares

BLP show that

- For a given set of $\boldsymbol{\theta}$ parameters, a unique vector $\boldsymbol{\delta}$ equates predicted market shares with observed market shares
- There is an iterative "contraction mapping" algorithm that recovers this unique vector of $\boldsymbol{\delta}$

Contraction Mapping Algorithm

We want to find the vector of product-market constant terms, $\boldsymbol{\delta}$, that equates predicted market share, $\widehat{S}_{jm}(\boldsymbol{\delta})$, with observed market share, S_{jm}, for all products in all markets

1. Begin with some initial product-market constant values, $\boldsymbol{\delta}^0$
2. Predict the market share for the current constant values, $\widehat{S}_{jm}(\boldsymbol{\delta}^s)$, for each product-market
3. Adjust each product-market constant term by comparing predicted and observed market share

$$\delta_{jm}^{s+1} = \delta_{jm}^{s} + \ln\left(\frac{S_{jm}}{\widehat{S}_{jm}(\boldsymbol{\delta}^s)}\right)$$

4. Repeat steps (2) and (3) until the algorithm converges to the set of product-market constants, $\widehat{\boldsymbol{\delta}}$

Market Share Inversion

For a given nonlinear parameter vector:

$$\theta = (\sigma_1, \ldots, \sigma_K)$$

BLP finds δ such that:

$$\boxed{s_j(\delta, \theta) = s_j^{obs}}$$

The inversion recovers the mean utility levels consistent with observed market shares.

BLP Contraction Mapping

BLP solves the share inversion using:

$$\delta_j^{(r+1)} = \delta_j^{(r)} + \log(s_j^{obs}) - \log(\hat{s}_j(\delta^{(r)}, \theta))$$

The iteration continues until:

$$||\delta^{(r+1)} - \delta^{(r)}|| < \epsilon$$

The fixed point gives:

$$\hat{\delta}(\theta)$$

Recovering Unobserved Product Quality

After obtaining δ_j:

$$\delta_j = x_j\bar{\beta} - \alpha p_j + \xi_j$$

Therefore:

$$\boxed{\xi_j = \delta_j - x_j\bar{\beta} + \alpha p_j}$$

The remaining term ξ_j captures unobserved product quality.

Price Endogeneity

Prices are endogenous because:

$$Cov(p_j, \xi_j) \neq 0$$

For example:

- high-quality cars have higher prices
- firms set prices based on unobserved demand factors

Therefore OLS estimation is biased.

Instrumental Variables

BLP uses instruments z_j satisfying:

$$\boxed{E[z_j \xi_j] = 0}$$

Typical instruments:

- characteristics of competing products
- number of competing products
- cost shifters

Estimation

The contraction mapping is an inner algorithm loop within the larger estimation loop

Two steps to estimate this model

1. Outer loop: search over $\boldsymbol{\theta}$ to optimize the estimation objective function
    1. Inner loop: use the contraction mapping to find $\boldsymbol{\delta}(\boldsymbol{\theta})$, the vector of product-market constant terms conditional on $\boldsymbol{\theta}$
    2. Use $\boldsymbol{\theta}$ and $\boldsymbol{\delta}(\boldsymbol{\theta})$ to simulate choice probabilities, $\check{P}_{njm}(\boldsymbol{\delta}(\boldsymbol{\theta}), \boldsymbol{\theta})$
    3. Use choice probabilities to calculate the estimation objective function
2. Estimate $\bar{\beta}$ by regressing δ_{jm} on $(p_{jm}, \boldsymbol{x}_{jm})$ with price instruments, $\boldsymbol{z}_{jm}$

We can estimate this model in two different ways

- MSL for step 1 and 2SLS for step 2
- MSM for steps 1 and 2 simultaneously

GMM Estimation

The moment condition:

$$g(\theta) = \frac{1}{J} \sum_j z_j \xi_j(\theta)$$

The BLP estimator:

$$\boxed{\hat{\theta} = \arg\min_{\theta} g(\theta)' W g(\theta)}$$

where:

$$\theta = (\alpha, \bar{\beta}, \sigma)$$

BLP Estimation Algorithm

1. Guess nonlinear parameters θ
2. Compute individual choice probabilities
3. Aggregate to predicted market shares
4. Solve contraction mapping for δ
5. Recover ξ:

$$\xi_j = \delta_j - x_j\bar{\beta} + \alpha p_j$$

6. Update parameters using GMM